

Huanhuan Ma

CONTACT INFORMATION

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EDUCATION

University of Chinese Academy of Sciences, Beijing, China

Sep 2021 – July 2024

M.S. in Artificial Intelligence, GPA: 3.51/4.00

- Advisors: Prof. Liang Wang, Prof. Qiang Liu
- Thesis: Explainable Automated Fact Verification Research
- Award: Ministry of Education - Huawei "Intelligent Base" Future Star 🏆

2022

Zhengzhou University, Zhengzhou, China

Sep 2016 – July 2020

B.S. in Software Engineering

RESEARCH
INTERESTS

My research primarily focuses on advancing Responsible **Generative Artificial Intelligence** (large language models), with an emphasis on:

- **Large model evaluation:** a) Traditional task-based evaluation [6][3], b) Psychological-based behavior evaluation of models [1]
- **Large model capability enhancement:** Targeted strengthening of model weaknesses through synthetic data generation

RESEARCH EXPERIENCE

Psychology Behavior Evaluation of LLM

December 2023 – Present

NC State University, GIC Lab, NC, USA

Intern Researcher, Mentor: **Dr. Dongkuan Xu**

- Leading a project to develop a more robust psychological behavior evaluation tool for assessing large language models (LLMs), with a manuscript submitted to ICLR 2025 [1]

Synthetic Data Generation

July 2024 – Present

Beijing Academy of Artificial Intelligence, (BAAI), Beijing, China

Research Intern, Mentors: **Dr. Li Du**, Hanyu Zhao, and Yiming Ju

- Contributed to generating high-quality synthetic data for model training and identifying model weaknesses

Fact-Checking

August 2022 - March 2024

Institute of Automation, Chinese Academy of Sciences, **CRIPAC**, Beijing, China

Master Student, Advisors: Prof. Liang Wang, Prof. Qiang Liu

- Proposed a novel explainable fact-checking dataset with textual explanations [3]
- Developed an innovative method for explainable multi-modal misinformation detection [4]
- Introduced a framework to address the weaknesses of large language models in fact-checking tasks [2]

LLM Knowledge Editing

July 2023 - March 2024

Institute of Automation, Chinese Academy of Sciences, **CRIPAC**, Beijing, China

Master Student

- Investigated entity and relational knowledge differences through knowledge localization [5]

PRE-PRINTS

- [1] **Huanhuan Ma**, Haisong Gong, Xiaoyuan Yi, Xing Xie, Dongkuan Xu. "Leveraging Implicit Sentiments: Enhancing Reliability and Validity in Psychological Trait Evaluation of LLMs" (**Under-review, ICLR**). [\[PDF\]](#),[\[OpenReview\]](#)
- [2] Haisong Gong, **Huanhuan Ma**, Qiang Liu, Shu Wu, Liang Wang. "Navigating the Noisy Crowd: Finding Key Information for Claim Verification." (**Arxiv, July 2024**). [\[PDF\]](#)

PEER-REVIEWED
PUBLICATIONS

- [3] **Huanhuan Ma**, Weizhi Xu, Yifan Wei, Liuji Chen, Liang Wang, Qiang Liu, Shu Wu, Liang Wang. "EX-FEVER: A Dataset for Multi-hop Explainable Fact Verification." *Findings of the Association for Computational Linguistics (ACL 2024 Findings)*. [\[PDF\]](#), [\[Code\]](#), [\[Poster\]](#), [\[Slides\]](#)
- [4] **Huanhuan Ma***, Jinghao Zhang*, Qiang Liu, Shu Wu, Liang Wang. "Interpretable Multi-modal Out-of-Context Detection with Soft Logic Regularization." *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2024 Oral)*. [\[PDF\]](#), [\[Slides\]](#)
- [5] Yifan Wei, Xiaoyan Yu, Yixuan Weng, **Huanhuan Ma**, Yuanzhe Zhang, Jun Zhao, Kang Liu. "Does Knowledge Localization Hold True? Surprising Differences Between Entity and Relation Perspectives in Language Models." *ACM International Conference on Information and Knowledge Management (CIKM 2024 Short)*. [\[PDF\]](#), [\[Code\]](#)
- [6] Yifan Wei, Yisong Su, **Huanhuan Ma**, Xiaoyan Yu, Fangyu Lei, Yuanzhe Zhang, Jun Zhao, Kang Liu. "MenatQA: A New Dataset for Testing the Temporal Comprehension and Reasoning Abilities of Large Language Models." *Findings of the Association for Computational Linguistics (EMNLP 2023 Findings)*. [\[PDF\]](#), [\[Code\]](#)
- [7] Liping Wang, Qiang Liu, **Huanhuan Ma**, Shu Wu, Liang Wang. "Multi-Cause Learning for Diagnosis Prediction." *International Conference on Data Mining and Big Data (DMBD 2022)*. [\[PDF\]](#)

PROJECTS

Awesome-LLM-based-Evaluators

[\[GitHub\]](#) ★23

- Curated a comprehensive list of LLM-based evaluators for various NLP tasks

INTERNSHIPS

Beijing Academy of Artificial Intelligence, BAAI, Beijing, China

July 2024 – Present

Research Intern, Mentors: **Dr. Li Du**, Hanyu Zhao, and Yiming Ju

- **Synthetic Data Generation**: Contributed to generating high-quality synthetic data for model training and identifying model weaknesses
- **Infinity-MM Project**: Contributed to the optimization of training data for enhancing large vision-language model

NC State University, GIC Lab, NC, USA

December 2023 – Present

Intern Researcher, Mentor: **Dr. Dongkuan Xu**

- Leading a project to develop a more robust psychological behavior evaluation tool for assessing large language models (LLMs), with a manuscript submitted to ICLR 2025 [\[1\]](#)

ACADEMIC
SERVICES

Program Committee

- The International Conference on Learning Representations (ICLR) 2025
- ACM International Conference on Information and Knowledge Management (CIKM) 2024
- Annual Conference on Neural Information Processing Systems (NeurIPS) Dataset & Benchmark track 2023

SKILLS

Language: Chinese (Native), English (Fluent)

Programming: Python, $\LaTeX$, SQL, HTML/CSS

Technologies: Git, Linux

Libraries: PyTorch, VLLM, LiteLLM, Faiss, SQLite